



Assessing Student Learning

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E.g., What do you want students to know (cognitive)?

E.g., What do you want students to be able to do (behavioral)?

Reflection: Your learning experiences

1. Think about your favorite class as an undergraduate or graduate student. What was the class?
2. How and when were you assessed in this class?

Discuss at your table:

- (a) Describe the type of assessment, and
- (b) When in the course was it used

Report Out & Reflect

Formative and Summative Assessment

- Formative: Information that students and/or instructors can use to improve at short timescales (e.g. clicker questions, "*minute papers*" or "*free-writing*", "*think-pair and share*")
- Summative: Information that assesses whether students and/or instructors have succeeded in meeting an objective (e.g. *final exams*, *research-based surveys*). Usually at the end of a course or sections of course.

(*An Introduction to Evidence-Based Undergraduate STEM Teaching* MOOC;
<https://www.stemteachingcourse.org/>)

Backward Design



Backward Design and Disciplinary Research

Teach ... like you do ... Research

*Learning Goals or Objectives

*Assessments

Data collected and provide feedback (data, evidence)

*Instructional Design and Class Activities



*Research Goals and Objectives,
Make Hypotheses

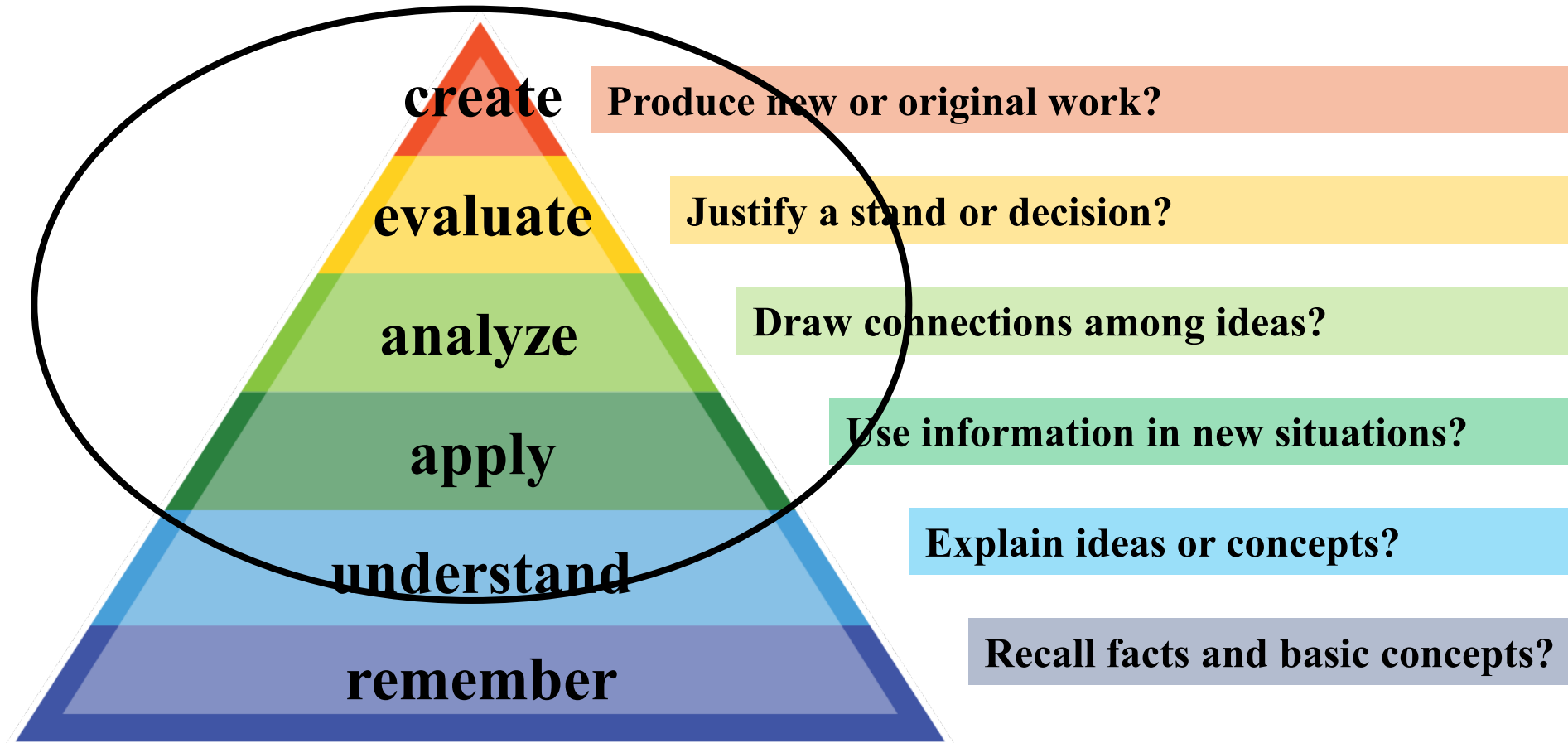
*Determine Acceptable Evidence
(data-quantitative, qualitative)

*Research Design, "Experiments"
(research activities)



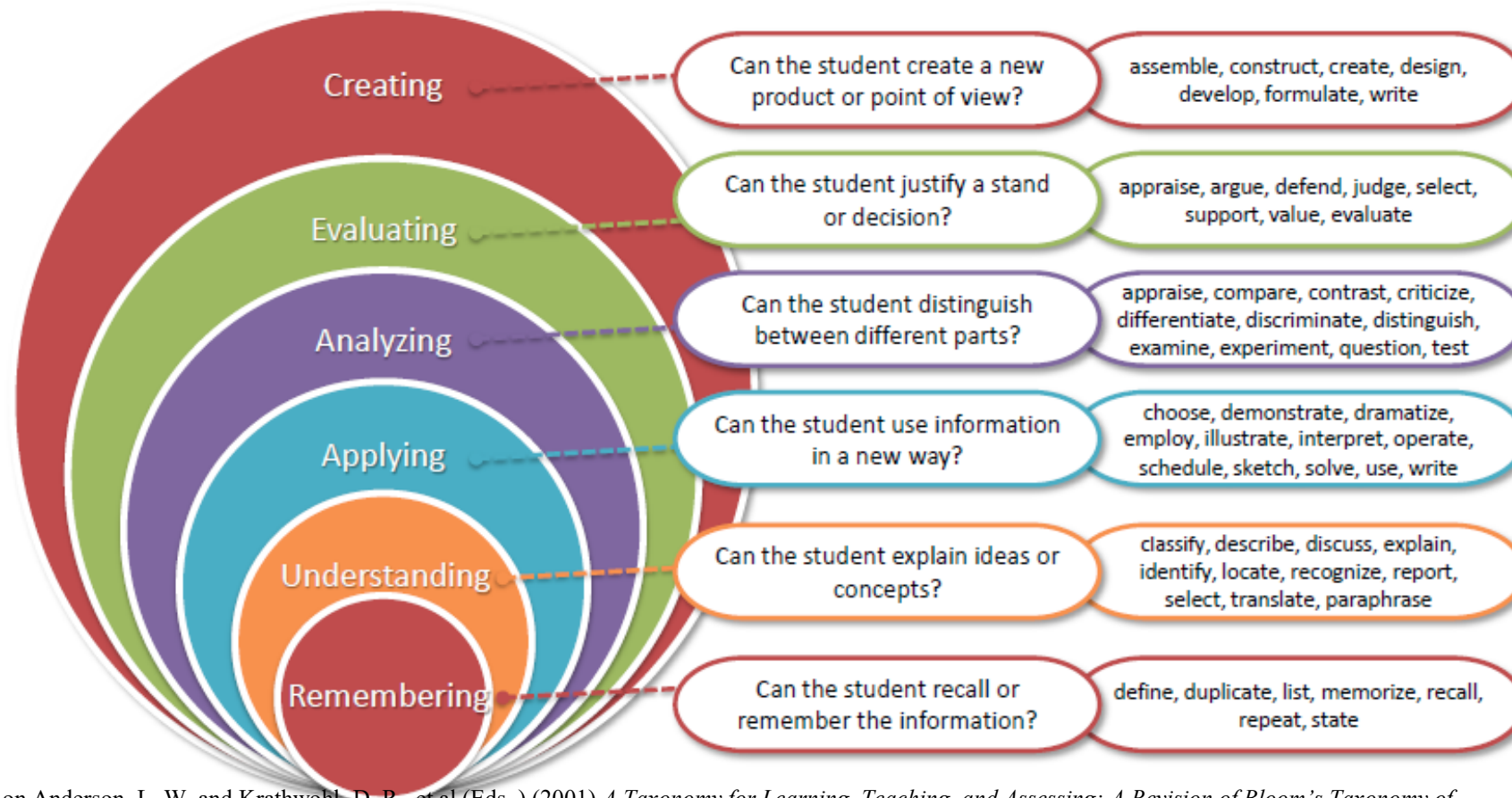
Bloom's Taxonomy

Can the student...



Bloom's Taxonomy (Revised)

"Bloom" your objectives!



Based on Anderson, L. W. and Krathwohl, D. R., et al (Eds..) (2001) *A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives*. Allyn & Bacon. Boston, MA

Image from <http://www.middleweb.com/wp-content/uploads/2015/03/Revised-Blooms-848.png>

Examples of Learning Objectives for Mentored Teaching Projects

- *Assess student's self-confidence and the effects of tutorials on student's overall performance in the course.*
- *Measure student performances after they have participated in a collaborative learning exercise and compare to students who have not*
- *Identify common misconceptions held by students associated with energy changes that occur during phase transitions*
- *Assess the relationship between students' self-reported reading completion and their understanding of the reading topics*

Observations?

E.g., Capstone Team Project

Misalignment

*What are your teaching and learning objectives for the team project? (Levels 1 & 2 – collect, identify, define)

*How will you evaluate teams (evaluation)?
(Levels 3 & 4 – apply, calculate, explain)
or
(Levels 5 & 6 – compare, discriminate)

*Constructing (i.e., teaching and mgt methods) an effective team project assignment?
(Levels 1 & 2 – collect, identify, define)

Misalignment = Students don't learn material + poor grades +
"I hate team projects."

Capstone Team Project

Alignment-Checked with “Blooming”

*What are your teaching and learning objectives for the team project? (Levels 1 & 2 – collect, identify, define, describe)

*How will you evaluate teams (evaluation)?
(Levels 1 & 2 – collect, identify, define, describe)

*Constructing (i.e., teaching and mgt methods) an effective team project assignment?
(Levels 1 & 2 – collect, identify, define, describe)

Alignment = All students benefit + high achievement +
learn professional skills + *“I learned a lot from this
team project.”*

Capstone Team Project

Alignment-Example 400-level Fisheries & Wildlife Class

*What are your teaching and learning objectives for the team project? *Describe the limitations of implementing the following clearcutting plan on wildlife communities and label (on map; and describe) how it could be improved (recommendations).*

*How will you assess teams?
Informal groups (2 or 3) – Groups should be able to describe 4 limitations and provide recommendations to improve conditions for wildlife (students turn in labeled map and descriptions)

*Constructing (i.e., teaching and mgt methods) an effective team project assignment?

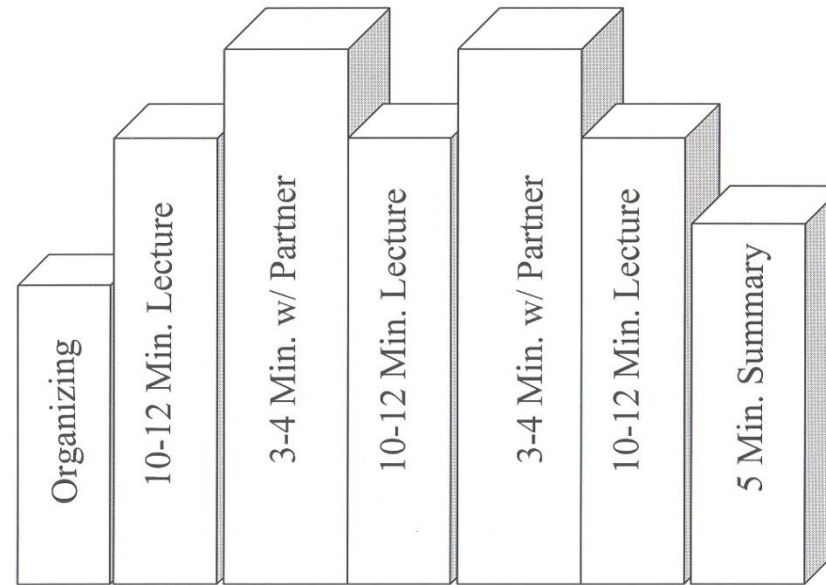
Students read material on forest mgt techniques prior to class; mini-lectures on even-aged and un-even-aged mgt methods

Alignment = All students benefit + high achievement +
learn professional skills + ***"I learned a lot from this team project."***

"Bookends" Model

Informal Cooperative Learning and
Effective Lecturing

(modified from K. Smith)



Reflection

2025 CCTI Folder

Go to ...

4 Assessing Student Learning Folder

- Open Reflections Worksheet Assessment
- Complete your reflection sheet

Instructions for Working Lunch

- Instructions for Micro Teaching (Rique)
 - Write threshold concept
 - Pair share (5 minutes)
 - Groups of 4—teach your concept
 - Assess your group's learning
- What is the concept they are teaching
- Based on Feedback teach to your group